

Exercise 0

Social Media Data Analysis

Basics

Python

We will use Python throughout this course, and you will also need to use it for your projects.

<https://www.python.org/>

If you are not yet familiar with Python, then please invest time getting up to speed as soon as possible. An [example tutorial](#) can be found in the documentation.

Basics

Package management

Beyond basic Python, we will use additional packages e.g. NumPy.

When collaborating as a team, you should make sure you are using the same versions of packages, as functionality can change between versions.

We recommend two package managers: **conda** or **uv**. As a team choose **one** of them.

<https://www.anaconda.com/download/success>

If you use conda: we recommend to install packages only with pip.

(<https://pypi.org/project/pip/>)

<https://github.com/astral-sh/uv>

Basics

Coding

When writing code, we often want more than just a text editor. There are many options with different features.

Jupyter notebooks enable interactive coding.

<https://jupyter-notebook.readthedocs.io/en/latest/>

There are also different computer programs that you can use called IDEs (Integrated Development Environment). These can help with highlighting errors in your code or making suggestions for code completion.

We recommend Visual Studio Code with python and jupyter extensions. Feel free to explore other options. <https://code.visualstudio.com/Download>

Version control

What is it? Why do we use it?

Version control enables the reversal of unwanted code changes and keeps track of progress.

It can be useful when coding alone, but it is especially important when **collaborating**. Using version control means that **multiple people** can work on the **code at the same time** and then handle **merging** all the changes together.

A record of each version is maintained, meaning that you can also **revert back** to previous versions if needed.

Version control

GitHub

One of the most popular methods of implementing version control is using GitHub.

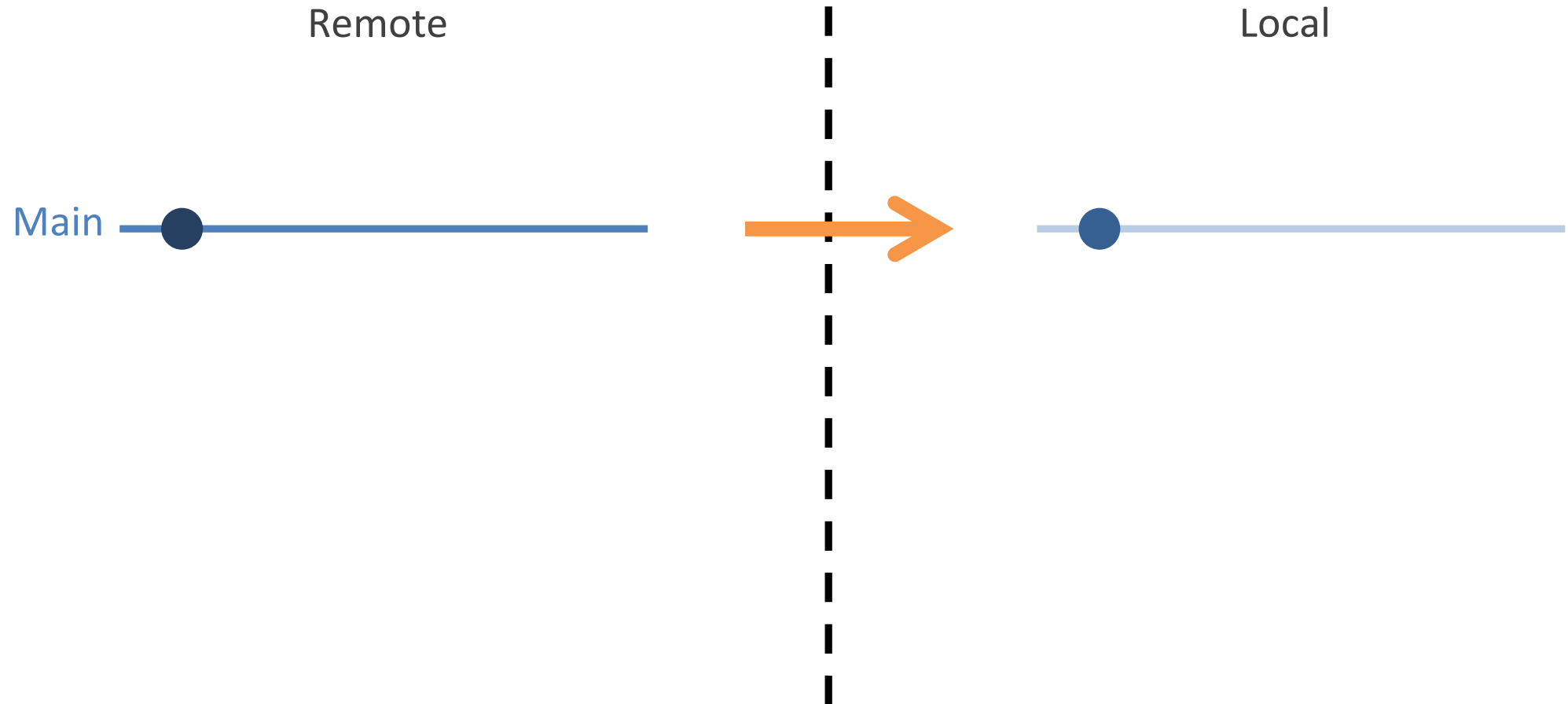
<https://docs.github.com/en/get-started/start-your-journey/about-github-and-git>

1. [Create an account](#)
2. [Set up Git](#) on your computer

Each code project is stored in a repository. There is a *remote* version of this online, but each user can also store a *local* version on their computer.

Version control

git pull



Version control

git branch new_branch_name

Remote

Main



Local



Feature

Version control

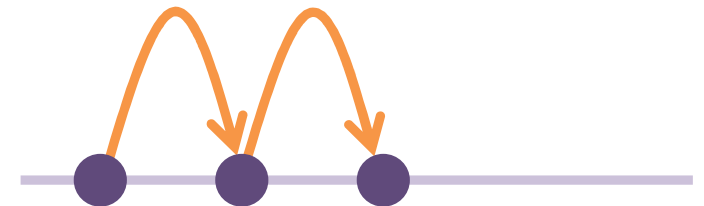
git commit -m "Added X"
Remote

Main ●—————

Local

●—————

Feature



Version control

git add
git push

Remote

Main



Local

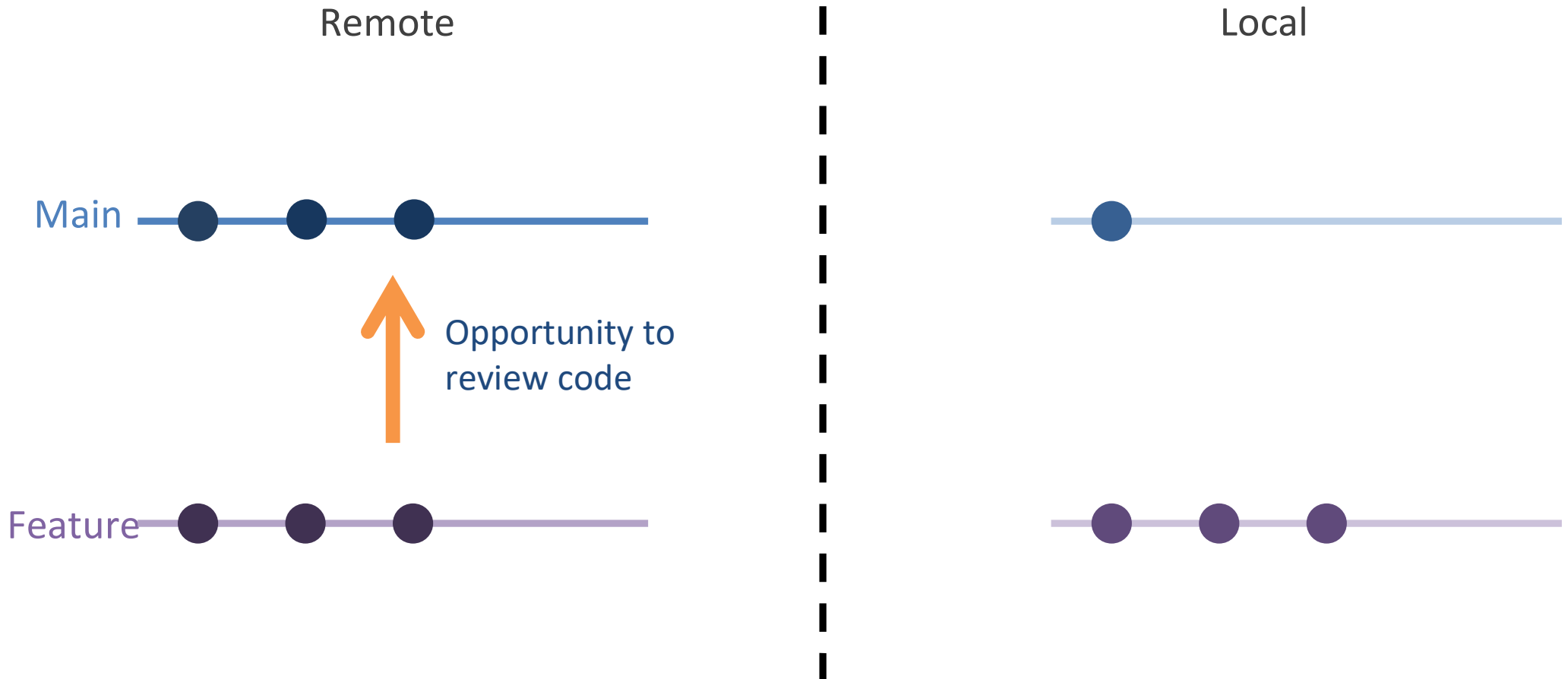


Feature



Version control

Pull request



Computing resources

Why do we need additional resources?

LLMs range in size but generally require more computing resources than a standard laptop.

They can be accessed via APIs if hosted elsewhere or run on GPUs.

Additional computing resources could be accessing models via an API or hosting them yourself on GPUs.

Google Colab

<https://colab.google/>

Uses Jupyter Notebooks and enables access to the Gemini API and to GPUs

Computing resources

BWUniCluster 3.0

https://wiki.bwhpc.de/e/BwUniCluster3.0/Getting_Started

High Performance Computing resource available through your membership of the university. After registration, you can request time using GPUs to run your experiments.

Getting access:

- A. Request entitlement via the university - [Request entitlement](#)
- B. Within 14 days of receiving entitlement, complete the questionnaire – [Questionnaire](#)
- C. Register for an account on the cluster - [Registration](#)

Can be used via [Slurm](#) or a [Jupyter notebook](#).

Currently, the BWUniCluster is down for maintenance. It should be up again on 18.02.2026